

Fraud & Anomaly Monitoring



Capstone Project
Business Analytics with
Gen & Agentic AI

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Part A- Problem Framing

1. Business Context

The Finance and Operations teams of the e-commerce platform have reported a rise in:

- Suspicious transactions (promo abuse, card testing, fake COD orders)
- High return and refund-related losses
- Unusually high discount usage and repeat cancellations

The company operates at scale, with over 4,600 orders, 100,000+ sessions, and multiple transactional layers including payments, shipments, refunds, and coupon usage. Manual monitoring is inefficient and reactive at this scale.

Leadership has requested an always-on fraud and anomaly monitoring system capable of:

1. Detecting major fraud patterns in near real-time
2. Ranking orders and users by risk for investigation
3. Estimating financial loss exposure
4. Recommending experiments and controls to reduce losses

Initial analysis indicates approximately ₹27L+ in potential refund and COD RTO exposure across the analysis period.

This project aims to build a structured, explainable, rule-based fraud monitoring framework using transactional and behavioral data.

2. Business Objective

Design an end-to-end fraud and anomaly monitoring system that:

- Identifies high-risk orders early
- Reduces financial losses from refunds and COD RTO (Return-to-Origin)
- Prioritizes cases for investigation
- Maintains stable revenue and conversion rates

The system must be explainable, scalable, and aligned with business decision-making needs.

3. North Star Metric

Monthly Fraud Loss Prevented (₹)

Justification:

- Directly measures the financial impact of the fraud monitoring system
- Provides a clear ROI view for leadership
- Captures prevention across multiple dimensions:
 - Refunds avoided
 - COD RTO losses reduced
 - High-discount abuse prevented
 - High-risk orders stopped before shipment

This metric is preferred over “reduction in high-risk orders shipped” because it quantifies actual financial benefit and aligns with leadership priorities.

4. Fraud / Anomaly Definition

An order is classified as “risky” if it exhibits one or more of the following signals:

- Unusually high discount percentage relative to gross amount
- Repeated coupon usage beyond expected behavior
- Multiple payment failures before success (card testing behavior)
- High order value or quantity compared to normal platform patterns
- Device ID or shipping pincode reuse across multiple accounts
- High refund probability signals
- Suspicious COD behavior with high RTO rate
- Clusters of users sharing device_id + pincode combinations

This rule-based and explainable definition ensures transparency and auditability.

5. Key Performance Indicators (KPIs)

Category	KPI	Definition / Formula
Loss & Refund Metrics	Refund Amount % of Gross Revenue	Total Refund Amount / Total Gross Revenue × 100
COD & Shipment Risk	COD RTO Rate (%)	COD Orders marked RTO / Total COD Orders × 100
COD & Shipment Risk	Estimated RTO Loss (₹)	Sum of net_amount for RTO orders
Discount & Coupon Monitoring	Coupon Usage Rate (%)	Orders with coupon / Total Orders × 100
Discount & Coupon Monitoring	Average Discount %	discount_amount / gross_amount × 100
Discount & Coupon Monitoring	Coupon Abuse Rate	High discount % + repeated coupon usage behavior
Payment Risk	Payment Failure Rate (%)	Failed payments / Total payment attempts × 100
Payment Risk	Orders with ≥3 Payment Failures (%)	Count of orders with 3+ failed payment attempts / Total Orders × 100
Fraud Monitoring	Suspicious Order Rate	High-risk flagged orders / Total Orders × 100

Category	KPI	Definition / Formula
Metrics	(%)	
Fraud Monitoring Metrics	Investigation Precision Proxy (%)	Flagged orders that actually result in refund or RTO / Total flagged orders $\times$ 100
Guardrails	Overall Conversion Rate (%)	Orders / Sessions $\times$ 100
Guardrails	Gross Revenue Trend (₹)	Sum of net_amount over time

6. Stakeholder Questions

This monitoring system should answer the following business-critical questions:

1. Which coupon is being abused the most and contributes to the highest refund loss? (linked to Coupon KPIs)
2. What percentage of COD orders result in RTO and what is the financial impact? (COD & Shipment KPIs)
3. Are payment failures clustered around specific devices or users (card testing pattern)? (Payment Risk KPIs)
4. Are new users exhibiting higher fraud signals compared to repeat customers? (Fraud Monitoring KPIs)
5. Which device IDs or shipping pincodes are reused across multiple risky accounts? (Fraud Monitoring KPIs)
6. What proportion of high-risk flagged orders actually lead to refund or RTO? (Investigation Precision Proxy)
7. If the top 10% highest-risk orders were blocked or manually reviewed, how much loss could be prevented? (North Star Metric)
8. Are fraud patterns increasing week-over-week? (Trend Analysis)

7. Success Measurement Framework

Success will be measured by:

- Reduction in monthly refund loss (₹)
- Reduction in COD RTO losses (₹)
- Reduction in high-risk orders shipped
- Stable or improved conversion rate
- Improved investigation precision over time

Goal: Reduce financial leakage without harming legitimate customer experience.

8. Strategic Positioning

The monitoring system will serve as a proactive risk intelligence layer integrated with order processing workflows, enabling real-time decision-making while maintaining operational efficiency.

Over time, the framework can evolve into a hybrid rule-based + machine learning risk engine for more sophisticated anomaly detection.